Phase 3, Block 0: Foundations of the finite-difference solution of the Black-Scholes PDE

Quant Finance Portfolio

1 Scope

This block sets up the mathematical and numerical scaffolding on which all of Phase 3 rests. It contains no pricer. Its deliverables are:

1. the logarithmic transformation that turns the Black–Scholes PDE into a constant-coefficient parabolic PDE;
2. the truncation of the spatial domain and the choice of artificial Dirichlet boundary conditions;
3. the discretisation grid and the standard centred finite-difference stencils;
4. the von Neumann stability analysis applied to the explicit (FTCS) and implicit (BTCS) schemes for the pure-diffusive part of the PDE.

The pricers themselves (FTCS, BTCS, Crank–Nicolson, PSOR, trinomial) appear in Blocks 1–5, all built on this foundation.

2 The Black–Scholes PDE

Under the risk-neutral measure $\mathbb{Q}$, the price $V(S,t)$ of a European option on a non-dividend-paying asset S_t following geometric Brownian motion

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t^{\mathbb{Q}}$$

satisfies the Black–Scholes partial differential equation

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV \;=\; 0, \qquad S > 0,\; t \in [0,T), \tag{1}$$

with terminal condition $V(S,T) = \phi(S)$, where ϕ is the option payoff. This is a backward parabolic equation in calendar time.

Direct discretisation of (1) on a uniform grid in S is unsatisfactory for two independent reasons. First, the coefficients $\frac{1}{2}\sigma^2 S^2$ and rS depend on the grid position, which complicates the von Neumann stability analysis: the standard analysis assumes constant coefficients. Second, a uniform grid in S resolves the lognormal-tailed distribution of S_T poorly, with too few nodes near $S = 0$ relative to the scale on which $V(S,t)$ varies.

Both problems are removed by a single change of variables.

3 Logarithmic transformation

Introduce

$$x \;=\; \ln\!\left(\frac{S}{K}\right), \qquad \tau \;=\; T - t. \tag{2}$$

The variable x is the log-moneyness; $\tau \in [0, T]$ is the time to maturity. Anchoring at the strike K (rather than using $\ln S$ directly) places $x = 0$ at the at-the-money point, which gives a canonical centring for the truncated domain.

Computing the derivatives by the chain rule, with $S = Ke^x$,

$$\frac{\partial V}{\partial t} \;=\; -\frac{\partial V}{\partial \tau}, \qquad \frac{\partial V}{\partial S} \;=\; \frac{1}{S}\frac{\partial V}{\partial x}, \qquad \frac{\partial^2 V}{\partial S^2} \;=\; \frac{1}{S^2}\left(\frac{\partial^2 V}{\partial x^2} - \frac{\partial V}{\partial x}\right).$$

Substituting into (1) and rearranging yields

$$\boxed{\frac{\partial V}{\partial \tau} \;=\; \frac{\sigma^2}{2}\frac{\partial^2 V}{\partial x^2} + \mu\,\frac{\partial V}{\partial x} - r\,V, \qquad \mu \;:=\; r - \frac{\sigma^2}{2}.} \tag{3}$$

This is a forward parabolic equation in τ on the unbounded domain $x \in \mathbb{R}$, with constant coefficients. The drift μ in (3) is precisely the drift of $\ln S_t$ under $\mathbb{Q}$, by Itô.

The initial condition becomes

$$V(x, 0) \;=\; \phi(Ke^x), \qquad x \in \mathbb{R}.$$

Remark 3.1. *The transformation* (2) *preserves the regularity of* V*. The non-smoothness of* ϕ *at the strike (the kink of* $(\cdot)^+$*) becomes a kink of the transformed initial condition at* $x = 0$*. This irregularity will be relevant in Block 3 when we discuss spurious oscillations in Crank–Nicolson.*

4 Structure of the transformed PDE

Equation (3) is a constant-coefficient *convection–diffusion–reaction* (CDR) equation:

$$V_\tau \;=\; D\,V_{xx} \;+\; a\,V_x \;+\; c\,V, \qquad D = \frac{\sigma^2}{2} > 0, \quad a = \mu \in \mathbb{R}, \quad c = -r \le 0.$$

Each of the three terms has a distinct numerical character.

- The *diffusive* term $D\,V_{xx}$ is dominant for typical financial parameters ($\sigma \in [0.1, 0.5]$, Δx small). It governs the most restrictive stability constraint on the explicit scheme of Block 1.
- The *convective* term $a\,V_x$ encodes directional drift. For $\mu > 0$ information flows toward $+x$; for $\mu < 0$ toward $-x$. With centred space differences (used throughout Phase 3) the contribution to the von Neumann factor is purely imaginary.
- The *reactive* term $cV = -rV$ is local in space and acts as a uniform exponential decay with rate r. It contributes only a multiplicative factor $1 - r\Delta\tau$ (or its implicit analogue) to the amplification factor and never destabilises the scheme.

Remark 4.1 (Why we keep all three terms together)**.** *A common simplification in textbooks is to apply a further change of variables* $V = e^{-r\tau}U$ *(kills the reactive term) and a Galilean shift in* x *(kills the convective term), reducing* (3) *to the pure heat equation* $U_\tau = D\,U_{xx}$*. This is mathematically clean but obscures the discrete structure: the boundary conditions and the payoff become time-dependent in non-trivial ways, and the relationship to the original* V *is harder to read off the code. Following the standard practice of Tavella & Randall [1], we discretise* (3) *as it stands.*

5 Truncation of the spatial domain

The unbounded domain $x \in \mathbb{R}$ has to be truncated to a finite interval $[x_{\min}, x_{\max}]$. The truncation introduces an artificial boundary; on it we will impose Dirichlet conditions whose values are read off the analytic asymptotics of V.

A standard choice is symmetric truncation centred at $x = 0$:

$$x_{\min} \;=\; -N_\sigma\,\sigma\sqrt{T}, \qquad x_{\max} \;=\; +N_\sigma\,\sigma\sqrt{T}, \tag{4}$$

with N_σ a small constant (typically 4 or 5). The justification is that the lognormal density of $S_T = Ke^{x_T}$ has $x_T \sim \mathcal{N}(\mu T,\, \sigma^2 T)$ under $\mathbb{Q}$, so the probability mass outside $[-N_\sigma\sigma\sqrt{T},\, +N_\sigma\sigma\sqrt{T}]$ is $O\big(e^{-N_\sigma^2/2}\big)$, which for $N_\sigma = 4$ is $\approx 6 \times 10^{-5}$ and decays super-polynomially in N_σ.

Remark 5.1. *The total error of any FD pricer is the sum of three contributions: the discretisation error (powers of Δx and $\Delta\tau$), the truncation error from the artificial boundary, and floating-point round-off. For $N_\sigma \geq 4$ the truncation contribution is typically below the discretisation error already at moderate $N \sim 100$ and never becomes the dominant error, so we treat it as negligible throughout. This separation of error sources will reappear in Block 3 when verifying the empirical convergence rate of CN.*

6 Boundary and initial conditions

European call

The terminal payoff $\phi(S) = (S - K)^+$ becomes, in x,

$$V(x, 0) \;=\; K\,(e^x - 1)^+, \qquad x \in [x_{\min}, x_{\max}]. \tag{5}$$

For the boundary conditions we use the well-known asymptotic limits. As $S \to 0$ the call goes to zero; as $S \to \infty$ it tends to the forward $S - Ke^{-r(T-t)}$. Translating to (x, τ),

$$V(x_{\min}, \tau) \;=\; 0, \tag{6}$$

$$V(x_{\max}, \tau) \;=\; K\big(e^{x_{\max}} - e^{-r\tau}\big)\,. \tag{7}$$

Both equalities hold in the limit $x_{\min} \to -\infty$, $x_{\max} \to +\infty$. On a truncated domain they are imposed as exact Dirichlet conditions; the small additional error this introduces is the truncation error discussed above.

European put

By analogous reasoning, the put payoff $\phi(S) = (K - S)^+$ becomes

$$V(x, 0) \;=\; K\,(1 - e^x)^+, \tag{8}$$

and the asymptotic boundary values are

$$V(x_{\min}, \tau) \;=\; K\big(e^{-r\tau} - e^{x_{\min}}\big)\,, \tag{9}$$

$$V(x_{\max}, \tau) \;=\; 0. \tag{10}$$

Equation (9) is a simple consequence of the martingale property: as $S \to 0$, $\mathbb{Q}\{S_T < K\} \to 1$ so $V(S, t) \to e^{-r(T-t)}\big(K - \mathbb{E}^{\mathbb{Q}}[S_T \mid S_t = S]\big) = Ke^{-r(T-t)} - S$, which is precisely $K(e^{-r\tau} - e^x)$.

Remark 6.1. *For Block 4 (American put) the boundary conditions* (9)–(10) *have to be replaced by the* intrinsic-value *conditions $V(x_{\min}, \tau) = K - Ke^{x_{\min}}$ (the option is exercised immediately for very low S) and the same $V(x_{\max}, \tau) = 0$. We will revisit this in Block 4.*

7 The discretisation grid

We discretise on a uniform tensor-product grid in (x, τ). Let

$$N \in \mathbb{N}, \quad M \in \mathbb{N}, \qquad \Delta x \;=\; \frac{x_{\max} - x_{\min}}{N}, \qquad \Delta\tau \;=\; \frac{T}{M}. \tag{11}$$

The spatial nodes are $x_j = x_{\min} + j\,\Delta x,\ j = 0, 1, \ldots, N$, giving $N+1$ nodes; the time levels are $\tau_n = n\,\Delta\tau,\ n = 0, 1, \ldots, M$, giving $M+1$ levels.

The discrete unknown is $V_j^n \approx V(x_j, \tau_n)$. The *interior nodes* are $j = 1, \ldots, N-1$ (a total of $N-1$ unknowns at each time level); the boundary nodes $j = 0$ and $j = N$ are fixed by the Dirichlet conditions of Section 6. At $\tau_0 = 0$ all nodes are fixed by the initial condition; the answer of interest is $V(0, T)$ if $S_0 = K$, or more generally the interpolant of V_j^M at $x = \ln(S_0/K)$.

The uniformity of the grid in x (not in S) is the consequence of the logarithmic transformation: a uniform grid in S would over-resolve the region near $S = 0$ and under-resolve large S. The transformation makes the natural sampling uniform in log-moneyness, which is also the natural scale for V to vary on.

8 Finite-difference stencils

Let f be a smooth function. Taylor expansion gives

$$f(x \pm h) \;=\; f(x) \pm h f'(x) + \frac{h^2}{2} f''(x) \pm \frac{h^3}{6} f'''(x) + O(h^4).$$

Combining the $+h$ and $-h$ expansions yields the centred stencils

$$f'(x) \;\approx\; \frac{f(x+h) - f(x-h)}{2h} + O(h^2), \tag{12}$$

$$f''(x) \;\approx\; \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} + O(h^2). \tag{13}$$

For the time derivative we use forward or backward Euler:

$$\partial_\tau V|_j^n \;\approx\; \frac{V_j^{n+1} - V_j^n}{\Delta\tau} + O(\Delta\tau), \qquad \text{(forward, used in FTCS)} \tag{14}$$

$$\partial_\tau V|_j^n \;\approx\; \frac{V_j^n - V_j^{n-1}}{\Delta\tau} + O(\Delta\tau). \qquad \text{(backward, used in BTCS)} \tag{15}$$

Centred-in-time differences (used in Crank–Nicolson, Block 3) achieve $O(\Delta\tau^2)$ at the cost of being implicit at both adjacent levels.

The label *FTCS* stands for *Forward-Time, Centred-Space* (combining (14) with (12)–(13) evaluated at level n); *BTCS* for *Backward-Time, Centred-Space* (combining (15) with the same spatial stencils, but evaluated at level $n+1$ from the perspective of the unknown).

9 Von Neumann stability analysis

Stability of a finite-difference scheme is the property that round-off errors do not grow unboundedly as the time-stepping proceeds. Together with consistency (the truncation error tends to zero as $\Delta x, \Delta\tau \to 0$), it implies convergence:

Theorem 9.1 (Lax equivalence, [4, Theorem 1.5.1])**.** *For a well-posed linear initial-value problem and a consistent finite-difference approximation, stability is necessary and sufficient for convergence.*

The criterion of stability we will use is the von Neumann criterion, which exploits the fact that on a uniform grid with constant coefficients the discrete Fourier modes $e^{ij\theta}$, $\theta \in [-\pi, \pi]$, diagonalise the spatial operators.

9.1 Setup

Write the round-off error at the level n as a single Fourier mode,

$$\varepsilon_j^n \;=\; g(\theta)^n\, e^{ij\theta}, \tag{16}$$

with $g(\theta) \in \mathbb{C}$ the (mode-dependent) *amplification factor*. Linearity of the scheme allows independent treatment of each θ. The scheme is *von Neumann stable* iff

$$|g(\theta)| \;\le\; 1 \qquad \text{for all } \theta \in [-\pi, \pi]. \tag{17}$$

9.2 FTCS for the pure-diffusive equation

Consider first the model problem $V_\tau = DV_{xx}$, ignoring the convective and reactive terms; we will reinstate them below. The FTCS discretisation is

$$\frac{V_j^{n+1} - V_j^n}{\Delta\tau} \;=\; D\,\frac{V_{j+1}^n - 2V_j^n + V_{j-1}^n}{\Delta x^2}.$$

Substituting the ansatz (16), dividing by $g^n e^{ij\theta}$, and using $e^{i\theta} + e^{-i\theta} = 2\cos\theta$:

$$g - 1 \;=\; \frac{D\,\Delta\tau}{\Delta x^2}\big(2\cos\theta - 2\big) \;=\; -4\,\alpha\,\sin^2\tfrac{\theta}{2},$$

where the dimensionless *Fourier number* is

$$\alpha \;:=\; D\,\frac{\Delta\tau}{\Delta x^2} \;=\; \frac{\sigma^2}{2}\cdot\frac{\Delta\tau}{\Delta x^2}. \tag{18}$$

Therefore

$$g(\theta) \;=\; 1 - 4\alpha\,\sin^2\tfrac{\theta}{2} \;\in\; \mathbb{R}.$$

The criterion (17) becomes $-1 \le g \le 1$. The upper bound is automatic since $\alpha, \sin^2 \ge 0$. The lower bound is tight at $\theta = \pi$ and gives

$$\boxed{\alpha \le \tfrac{1}{2}} \quad\Longleftrightarrow\quad \frac{\sigma^2\Delta\tau}{\Delta x^2} \le 1. \tag{19}$$

(19) is the *CFL condition* for the heat equation, in honour of Courant, Friedrichs and Lewy. It says that an explicit diffusive scheme cannot propagate the heat by more than half a grid spacing per time step. The constant 1/2 is sharp.

Remark 9.2. *The CFL constraint is severe in practice. For $\sigma = 0.2$, $\Delta x = 0.05$, the largest stable $\Delta\tau$ is $\alpha\Delta x^2/D \approx \frac{0.5\cdot 0.0025}{0.02} = 0.0625$, so a year requires at least $T/\Delta\tau = 16$ time steps. Halving Δx requires* quadrupling *the number of time steps. This $O(\Delta x^{-2})$ scaling of the time-step count is the principal limitation of FTCS and the motivation for the implicit schemes in Block 2 and beyond.*

9.3 BTCS for the pure-diffusive equation

The implicit scheme evaluates the spatial stencil at level $n+1$:

$$\frac{V_j^{n+1} - V_j^n}{\Delta\tau} \;=\; D\,\frac{V_{j+1}^{n+1} - 2V_j^{n+1} + V_{j-1}^{n+1}}{\Delta x^2}.$$

Substituting (16) gives

$$g - 1 \;=\; -4\alpha\, g\,\sin^2\tfrac{\theta}{2} \;\Longrightarrow\; g(\theta) \;=\; \frac{1}{1 + 4\alpha\sin^2(\theta/2)}.$$

Since $\alpha > 0$, the denominator is at least 1 for every θ, so $g \in (0, 1]$. The criterion is satisfied for any $\alpha > 0$:

Proposition 9.3 (Unconditional stability of BTCS)**.** *The BTCS scheme for $V_\tau = DV_{xx}$ is von Neumann stable for any $\Delta\tau > 0,\ \Delta x > 0$.*

This is the central trade-off of Phase 3. FTCS is cheap per step (matrix-free) but has the CFL constraint; BTCS allows arbitrary $\Delta\tau$ but requires solving a linear system at each step. Block 2 will pay the cost of that linear solve via the Thomas algorithm.

9.4 Convective term: an aside

Reinstating the convective term aV_x with centred-space differences in FTCS, the amplification factor becomes complex,

$$g(\theta) \;=\; 1 - 4\alpha \sin^2\tfrac{\theta}{2} + i\,\nu\,\sin\theta, \qquad \nu \;:=\; a\,\frac{\Delta\tau}{2\,\Delta x},$$

where ν is the *Courant number* (advective). Imposing $|g|^2 \le 1$ yields a mixed condition coupling α and ν; algebraically, after some manipulation (see [4, §3.5]),

$$4\alpha^2 \;\ge\; \nu^2 \quad \text{and} \quad \alpha \le \tfrac{1}{2}.$$

For Black–Scholes parameters the relevant figure of merit is the *grid Péclet number* $\mathrm{Pe}_h = |a|\Delta x/(2D) = |\mu|\Delta x/\sigma^2$. For $\sigma = 0.2$, $|\mu| \le 0.1$, $\Delta x \le 0.1$, $\mathrm{Pe}_h \le 0.25$, so the convective contribution is sub-dominant and the diffusive bound (19) governs in practice. The reactive term $-rV$, being local in space, contributes only a multiplicative factor $1 - r\Delta\tau$ to g and does not affect stability for $\Delta\tau \le 1/r$, which is always satisfied.

9.5 Summary: two control numbers

The discrete dynamics of the entire phase is controlled by two dimensionless numbers,

$$\alpha \;=\; \frac{\sigma^2}{2}\cdot\frac{\Delta\tau}{\Delta x^2}, \qquad \nu \;=\; \frac{|\mu|}{2}\cdot\frac{\Delta\tau}{\Delta x}. \tag{20}$$

FTCS (Block 1) is stable iff $\alpha \le 1/2$ and the convective contribution does not violate the mixed bound; in the parameter range of this project this reduces effectively to $\alpha \le 1/2$. BTCS (Block 2) and Crank–Nicolson (Block 3) are unconditionally stable in α. As we will see in Block 5, the Kamrad–Ritchken trinomial tree maps to FTCS under a specific reparametrisation; its non-negative-probability condition $\lambda^2 \ge 1/\alpha$ is the discrete-tree avatar of the CFL constraint (19).

10 Implementation outline

The Phase 3 Block 0 implementation provides four ingredients:

1. `Grid` (Python dataclass / C++ struct): an immutable container holding N, M, T, σ, r, K, $x_{\min}$, $x_{\max}$, Δx, $\Delta\tau$, the arrays $\{x_j\}$ and $\{\tau_n\}$.
2. `build_grid`: a constructor that picks $x_{\min}, x_{\max}$ from $\pm N_\sigma\sigma\sqrt{T}$ (default $N_\sigma = 4$) and assembles the grid.
3. Initial-condition functions `call_initial_condition`, `put_initial_condition` returning the payoff vector at $\tau = 0$.
4. Boundary functions `call_boundary_lower`, `call_boundary_upper`, `put_boundary_lower`, `put_boundary_upper` returning Dirichlet values for every τ_n.
5. Stability-number utilities `fourier_number` and `courant_number` for diagnostic use by the explicit pricer in Block 1.

This module imports nothing from outside `numpy` and the standard library. It contains no pricer.

References

References

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